>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

SENSITIVITY ANALYSIS BY CHANGING STRUCTURAL DAMPER STIFFNESS AND DAMPING RATIO USING OPENSEES

THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)

Suppose we have an existing steel or concrete structure and wish to enhance its seismic performance by adding supplemental dampers. The goal is to increase the overall effective damping of the structural system through the contribution of these devices. In this type of study a sensitivity analysis is carried out by varying key damper parameters; such as the elastic stiffness of the damper and its viscous damping coefficient in order to evaluate the resulting total damping of the structure. This total damping combines the inherent damping of the original structure and the additional damping provided by the dampers.

Energy quantities and the damping formula

- **Dissipated energy** E_d – the area enclosed by the hysteresis loop. This is the net work done per cycle and represents the energy that must be removed from the system to return it to the original state. Physically, it corresponds to plastic yielding, friction, cracking, or other irreversible mechanisms.
- **Maximum strain energy** E_s – defined here as

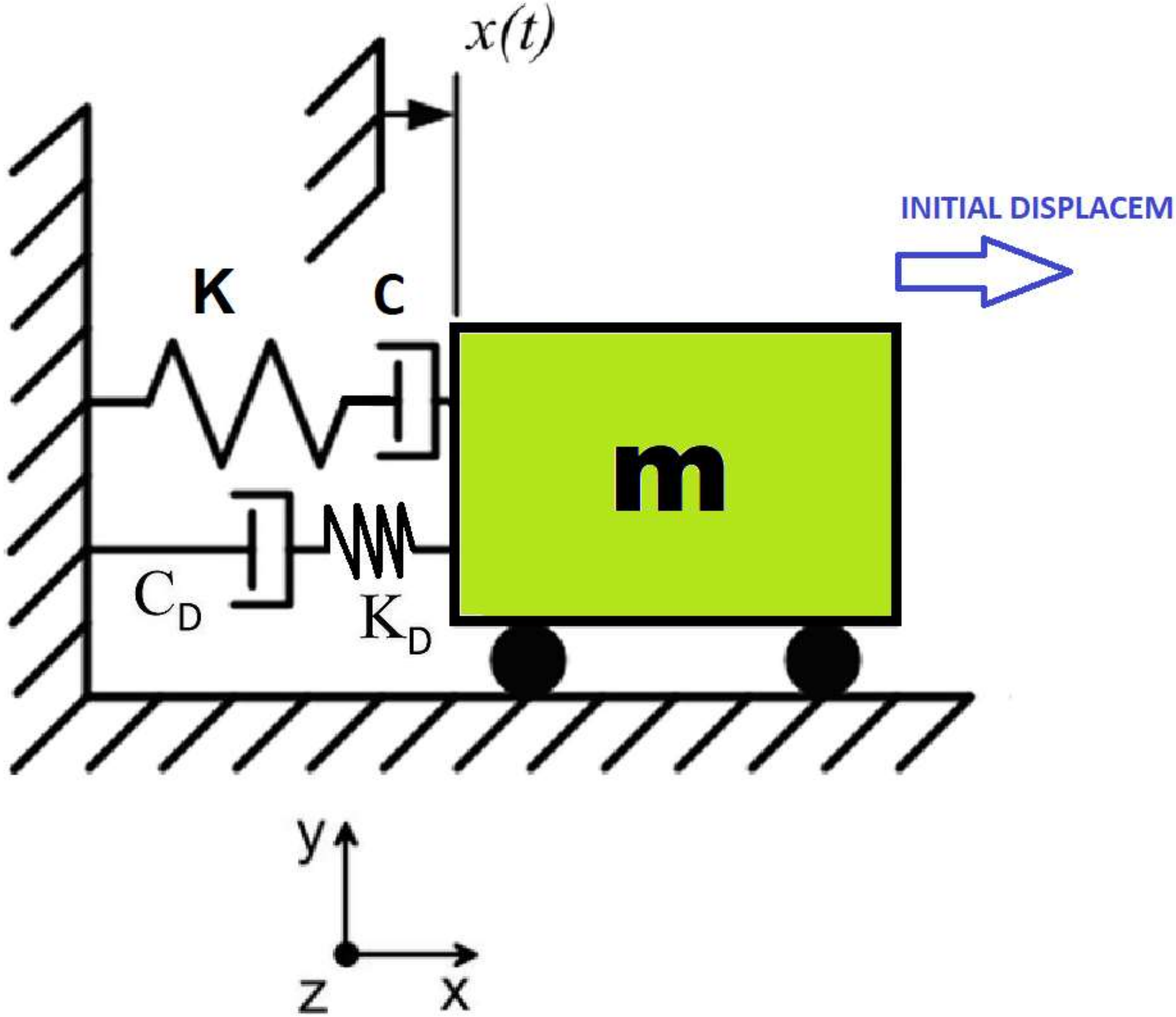
$$E_s = \frac{1}{2} F_{\max} u_{\max}$$

where $u_{\max} = \max |u|$ is the peak absolute displacement in the loop and $F_{\max}$ is the absolute value of the force **at that same displacement**. This definition assumes a **secant stiffness** $k_{sec} = F_{\max}/u_{\max}$ and represents the elastic strain energy stored at peak displacement if the system were linear with that stiffness.

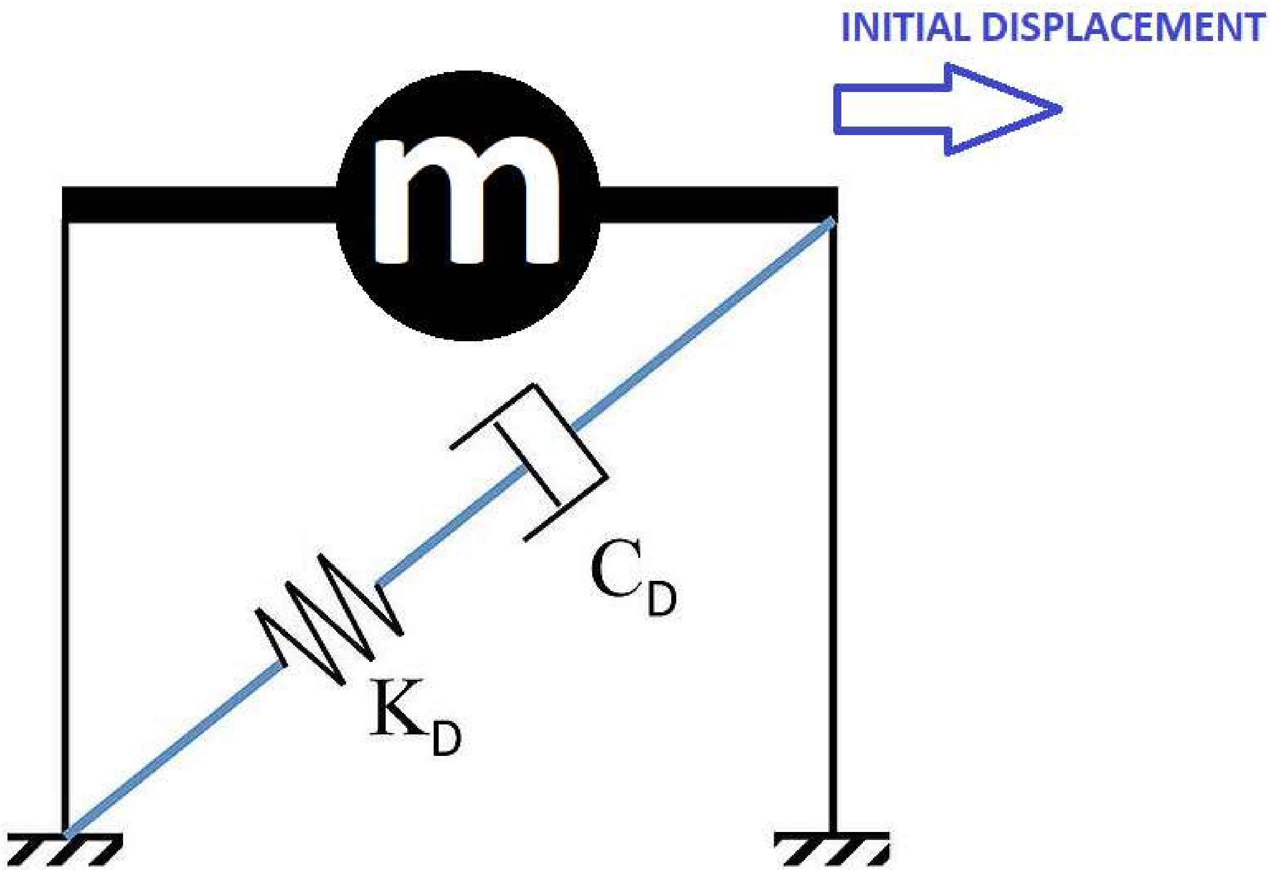
The equivalent viscous damping ratio is then:

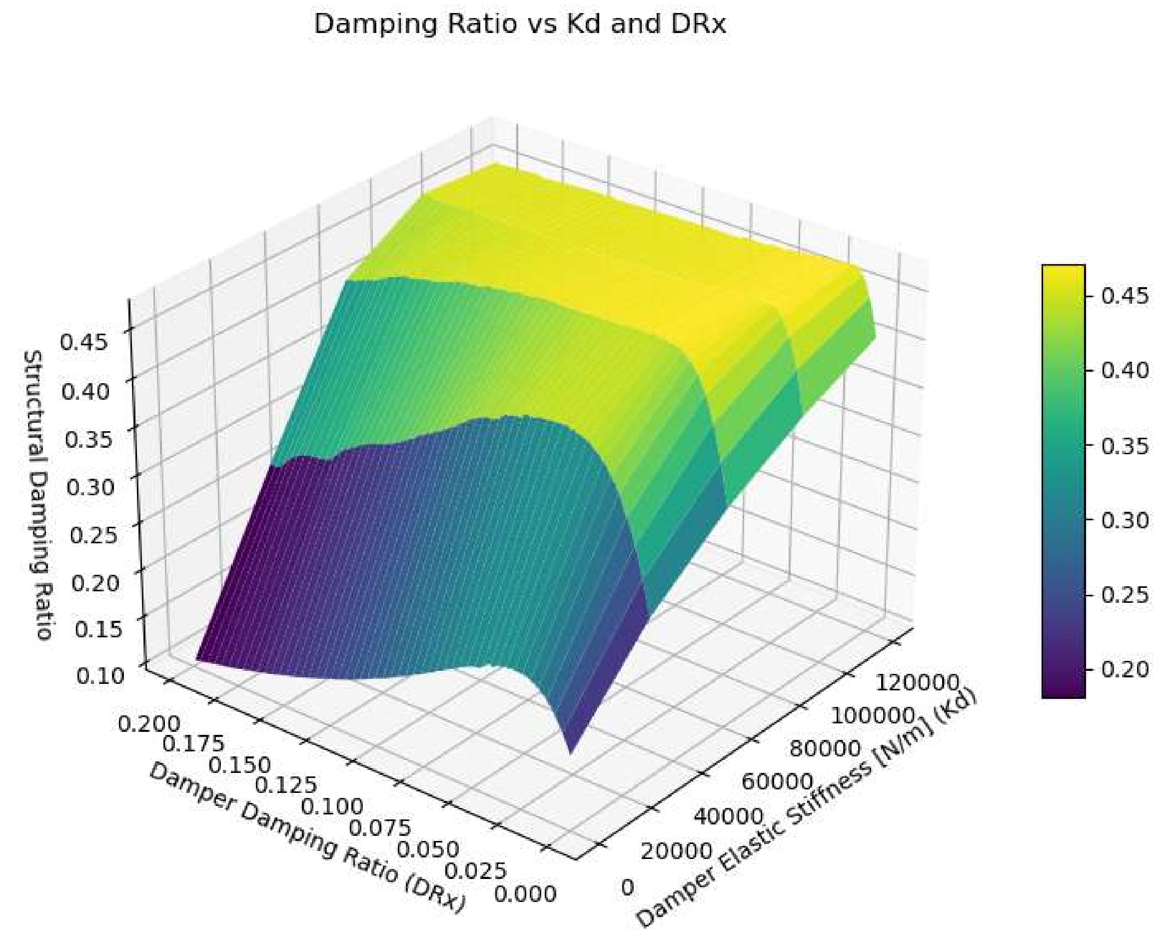
$$\xi_{eq} = \frac{100 \, E_d}{4\pi \, E_s} \quad [\%]$$

This stems from equating E_d to the energy dissipated per cycle by a linear viscous damper, $E_d = \pi c \omega u_{\max}^2$, while $E_s = \frac{1}{2} k u_{\max}^2$; substituting $\xi = c/(2m\omega)$ and $\omega^2 = k/m$ yields the familiar expression $\xi = E_d/(4\pi E_s)$.

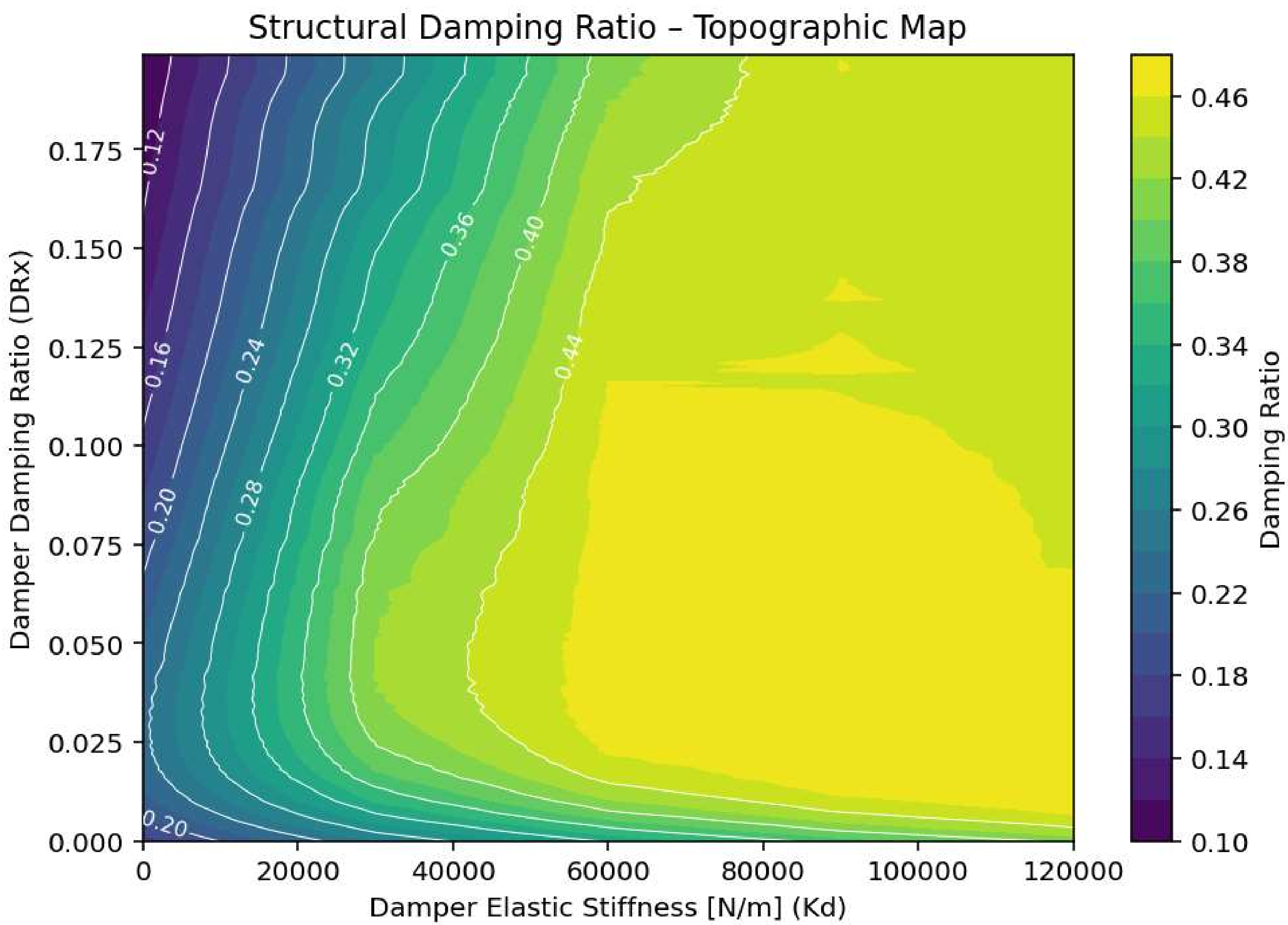


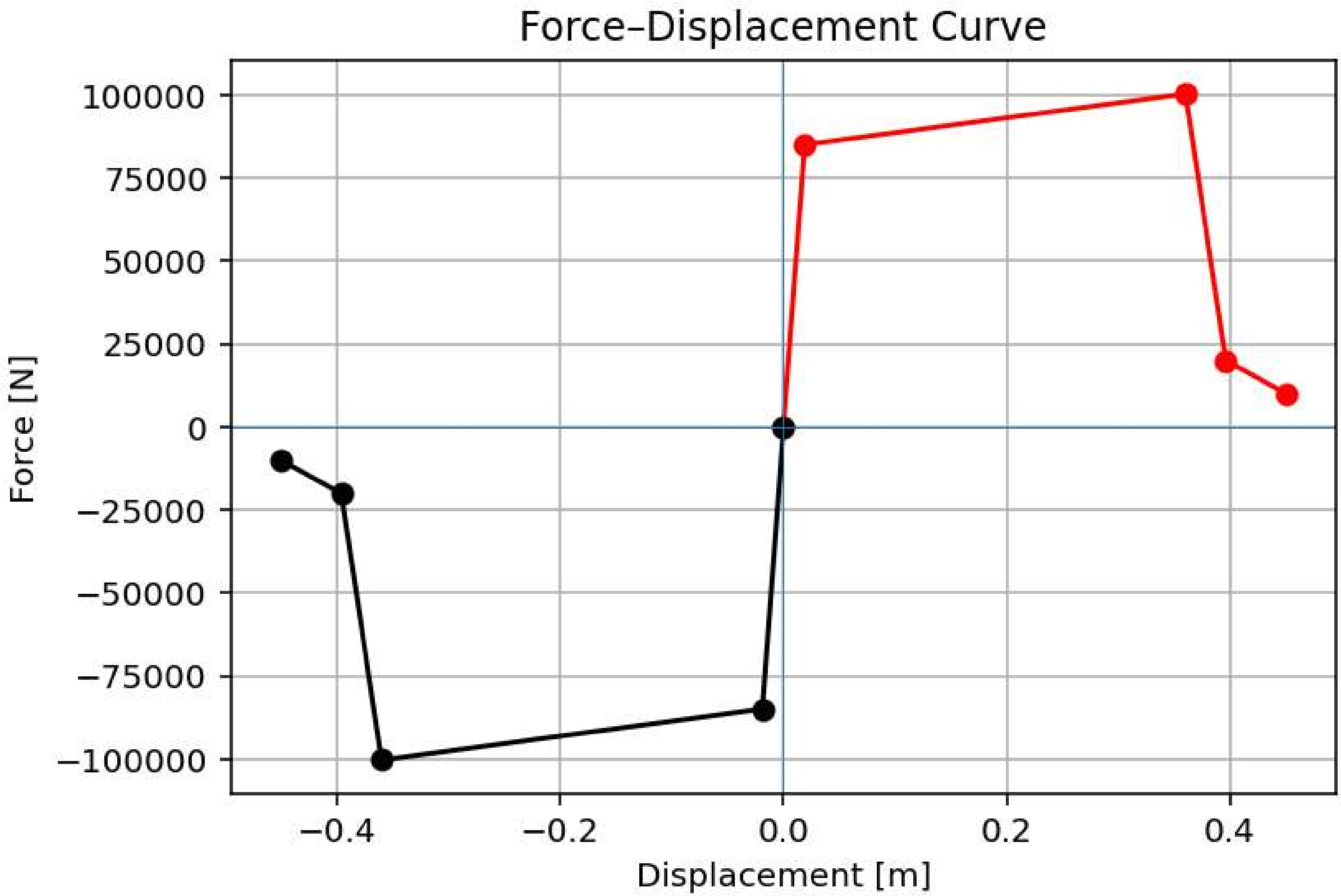
x(t)
INITIAL DISPLACEM
K
C
m
C_D
K_D
y
z
x


INITIAL DISPLACEMENT
m
C
D
K
D

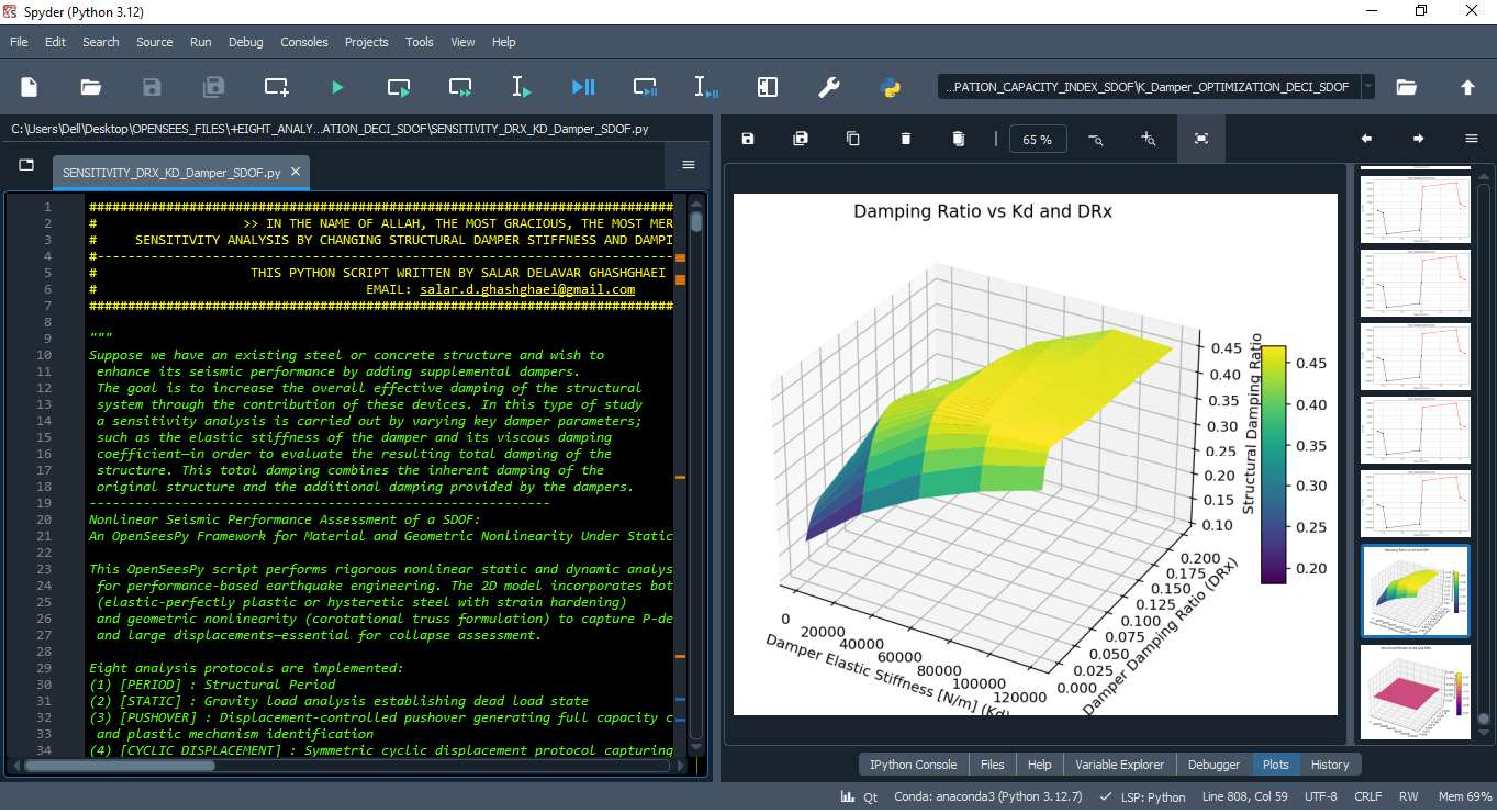


Damping Ratio vs Kd and DRx
Structural Damping Ratio
0.10
0.15
0.20
0.25
0.30
0.35
0.40
0.45
Damper Damping Ratio (DRx)
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
Damper Elastic Stiffness [N/m] (Kd)
0
20000
40000
60000
80000
100000
120000
0.20
0.25
0.30
0.35
0.40
0.45


Structural Damping Ratio – Topographic Map
Damper Damping Ratio (DRx)
Damper Elastic Stiffness [N/m] (Kd)
Damping Ratio
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0
20000
40000
60000
80000
100000
120000
0.10
0.14
0.18
0.22
0.26
0.30
0.34
0.38
0.42
0.46
0.12
0.16
0.20
0.24
0.28
0.32
0.36
0.40
0.44
0.20



Force–Displacement Curve
100000
75000
50000
25000
0
−25000
−50000
−75000
−100000
Force [N]
−0.4
−0.2
0.0
0.2
0.4
Displacement [m]


Spyder (Python 3.12)
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SENSITIVITY_DRX_KD_Damper_SDOF.py
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SENSITIVITY ANALYSIS BY CHANGING STRUCTURAL DAMPER STIFFNESS AND DAMPI
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THIS PYTHON SCRIPT WRITTEN BY SALAR DELAVAR GHASHGHAEI
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"""
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 enhance its seismic performance by adding supplemental dampers.
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 system through the contribution of these devices. In this type of study
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 original structure and the additional damping provided by the dampers.

Nonlinear Seismic Performance Assessment of a SDOF:
An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static
This OpenSeesPy script performs rigorous nonlinear static and dynamic analys
 for performance-based earthquake engineering. The 2D model incorporates bot
 (elastic-perfectly plastic or hysteretic steel with strain hardening)
 and geometric nonlinearity (corotational truss formulation) to capture P-de
 and large displacements—essential for collapse assessment.
Eight analysis protocols are implemented:
(1) [PERIOD] : Structural Period
(2) [STATIC] : Gravity load analysis establishing dead load state
(3) [PUSHOVER] : Displacement-controlled pushover generating full capacity c
 and plastic mechanism identification
(4) [CYCLIC DISPLACEMENT] : Symmetric cyclic displacement protocol capturing
65 %
Damping Ratio vs Kd and DRx
Damper Elastic Stiffness [N/m] (Kd)
Damper Damping Ratio (DRx)
Structural Damping Ratio
0 20000 40000 60000 80000 100000 120000
0.000 0.025 0.050 0.075 0.100 0.125 0.150 0.175 0.200
0.10 0.15 0.20 0.25 0.30 0.35 0.40 0.45
0.20 0.25 0.30 0.35 0.40 0.45
IPython Console Files Help Variable Explorer Debugger Plots History
Qt Conda: anaconda3 (Python 3.12.7) LSP: Python Line 808, Col 59 UTF-8 CRLF RW Mem 69%